Solution:
By the half-angle identity, we have

$$\cos\phi = \frac{1-\tan^2\frac{\phi}{2}}{1+\tan^2\frac{\phi}{2}} = \frac{1-\left(\frac{1+a}{1-a}\right)\tan^2\frac{\theta}{2}}{1+\left(\frac{1+a}{1-a}\right)\tan^2\frac{\theta}{2}}$$

$$= \frac{\left(\frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}\right)-a}{1-a\left(\frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}\right)} = \frac{\cos\theta-a}{1-a\cos\theta}.$$

Example 3.31 Find the value of $\sqrt{3}\,\text{cosec}\,20^\circ - \sec 20^\circ$

Solution:

$$\text{We have } \sqrt{3}\,\text{cosec}\,20^\circ - \sec 20^\circ = \frac{\sqrt{3}}{\sin 20^\circ} - \frac{1}{\cos 20^\circ} = 4\left[\frac{\frac{\sqrt{3}}{2}\cos 20^\circ - \frac{1}{2}\sin 20^\circ}{2\sin 20^\circ\cos 20^\circ}\right]$$

$$= 4\left[\frac{\sin 60^\circ\cos 20^\circ - \cos 60^\circ\sin 20^\circ}{\sin 40^\circ}\right] = 4.$$

Example 3.32 Prove that $\cos A\cos 2A\cos 2^2A\cos 2^3A\ldots\cos 2^{n-1}A = \frac{\sin 2^nA}{2^n\sin A}$

Solution:

$$\text{L.H.S.} = \cos A\cos 2A\cos 2^2A\cos 2^3A\ldots\cos 2^{n-1}A$$

$$= \frac{1}{2\sin A}2\sin A\cos A\cos 2A\cos 2^2A\cos 2^3A\ldots\cos 2^{n-1}A$$

$$= \frac{1}{2\sin A}\sin 2A\cos 2A\cos 2^2A\cos 2^3A\ldots\cos 2^{n-1}A$$

$$= \frac{1}{2^2\sin A}\sin 4A\cos 2^2A\cos 2^3A\ldots\cos 2^{n-1}A$$

Continuing the process, we get

$$= \frac{\sin 2^nA}{2^n\sin A} = \text{R.H.S}$$

Exercise - 3.5

1. Find the value of $\cos 2A$, A lies in the first quadrant, when
   (i) $\cos A = \frac{15}{17}$ (ii) $\sin A = \frac{4}{5}$ (iii) $\tan A = \frac{16}{63}$.